

ETHICAL CONCERN IN DATA COLLECTION

(SOC 2101 - Society, Environment and Engineering Ethics)

1. Ethical Considerations in Data Collection

Ethical considerations refer to the moral and responsible practices involved in how data is:

- collected
- stored
- shared

They ensure that data collection does not harm individuals and respects their rights, privacy, and autonomy.

No Harm

Key ethical concerns include:

- informed consent
- safe data storage
- permission for data use and sharing

Data → Collected → Stored → Shared

2. Informed Consent

Informed consent refers to written permission given by a person to participate in any activity where private data or personal information may be collected.

A proper informed consent document clearly explains:

- the purpose of data collection
- why the data is being collected
- from whom the data is collected
- how the data will be collected
- how it will be stored
- how long it will be retained
- who will have access to the data
- Informed consent ensures that participation is voluntary and aware, not forced or hidden.

★ Common Mistake:
Treating consent
as "implied"

☒ Voluntary



3. Confidentiality and Anonymity

Confidential data:	Anonymous data:
• Data linked to a specific individual • Kept private and protected Example: medical records, service records	• Data that cannot be traced back to an individual • No personal identifiers are retained

Participants must clearly know:

- whether their data is confidential or anonymous
- how their identity will be protected

💡 Memory Trick:

Confidential =
Connected to name,
Anonymous =
Absolutely no name

4. Clear Communication and Data Sharing

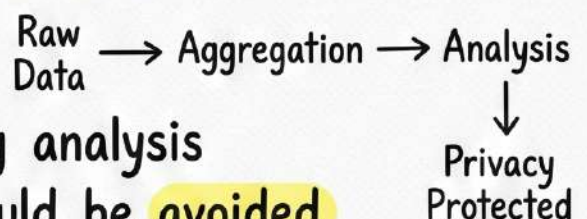
Ethical responsibility does not end with data collection.

Researchers must clearly communicate:

- how collected data will be used
- how it will be shared
- whether it will be published or analyzed

To protect privacy:

- data should be aggregated during analysis
- individual-level identification should be avoided



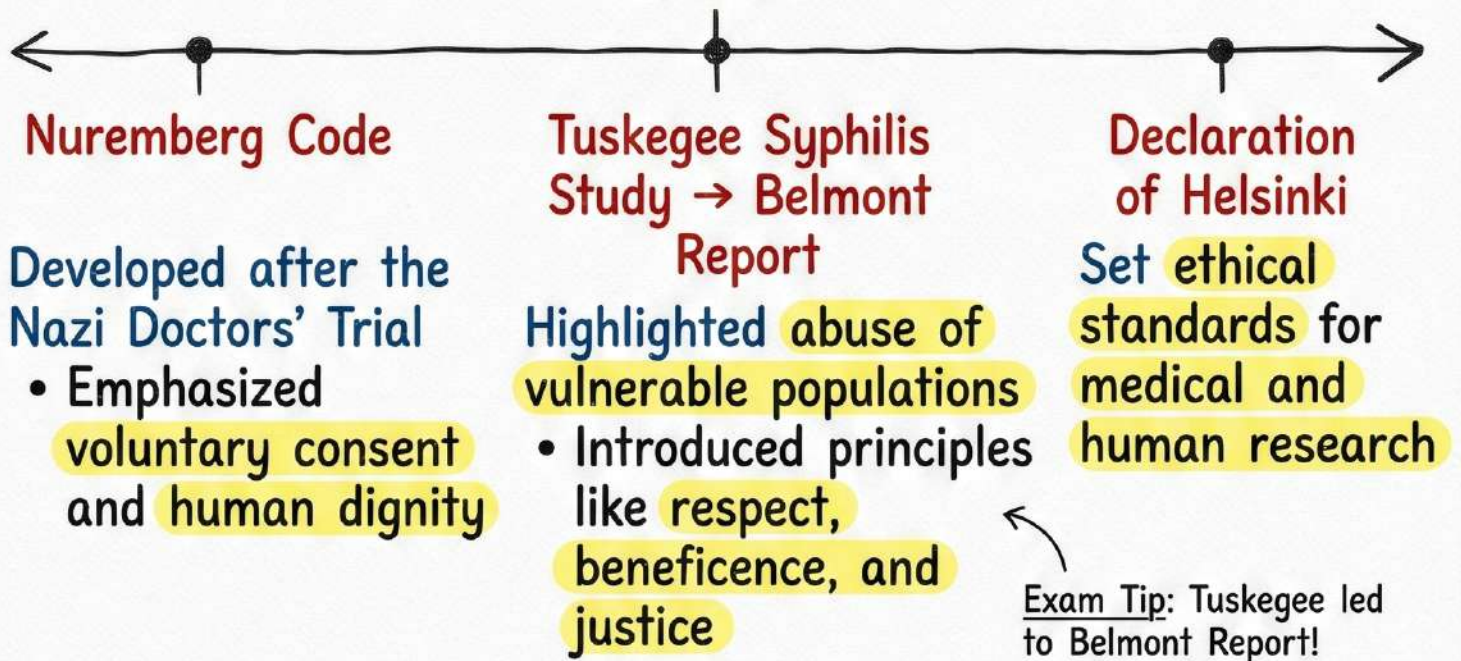
Participants should not be surprised later about how their data was used.



5. Ethical Issues in Human Experimentation (Historical Context)

Past unethical experiments shaped modern data ethics.

Timeline



These cases show why strict ethical rules are necessary in data collection involving humans.

6. Data Protection Landscape

HIPAA (1996)

Designed to protect personal health information.

- Medical data shared only on a "need-to-know" basis.
- Data can be shared only with patient consent, except in specific cases:
 - gunshot or stab wounds
 - crime-related injuries
 - abuse cases
 - infectious diseases

Updated in 2013:

- patients can access electronic records
- genetic data included



Data Protection Landscape

↳ GDPR (General Data Protection Regulation)

- GDPR provides strong protection of personal data.
- Data processing is allowed only if:
 - the individual has given explicit consent, or
 - it is required to fulfill a contract, or
 - it is necessary for legal obligations, or
 - it protects vital interests, or
 - it serves public interest or official authority
- GDPR emphasizes:
 - user control
 - transparency
 - accountability

7. Common Exam Mistakes (Avoid These)

- Treating consent as "implied"
- Mixing up confidentiality and anonymity
- Forgetting data storage and access control
- Ignoring data-sharing ethics
- Writing GDPR/HIPAA names without explaining purpose

8. Keywords to Remember (Memory Hack)

Think **I-C-A-S-H-G**



9. Likely Question Patterns

- Explain informed consent and its importance
- Differentiate confidentiality and anonymity
- What ethical concerns exist in data sharing?
- Why are historical experiments important in data ethics?
- Role of HIPAA / GDPR in data protection



Good Luck!
